

UNIT 2 - Class Activity (24.07.2026)

Employee Salary Data Visualization Using R

1. DATASET

```
data <- read.csv(file.choose())  
head(data)  
str(data)
```

2. AMOUNTS

Visualization 1: Bar Chart

Title: Average Monthly Salary by Department

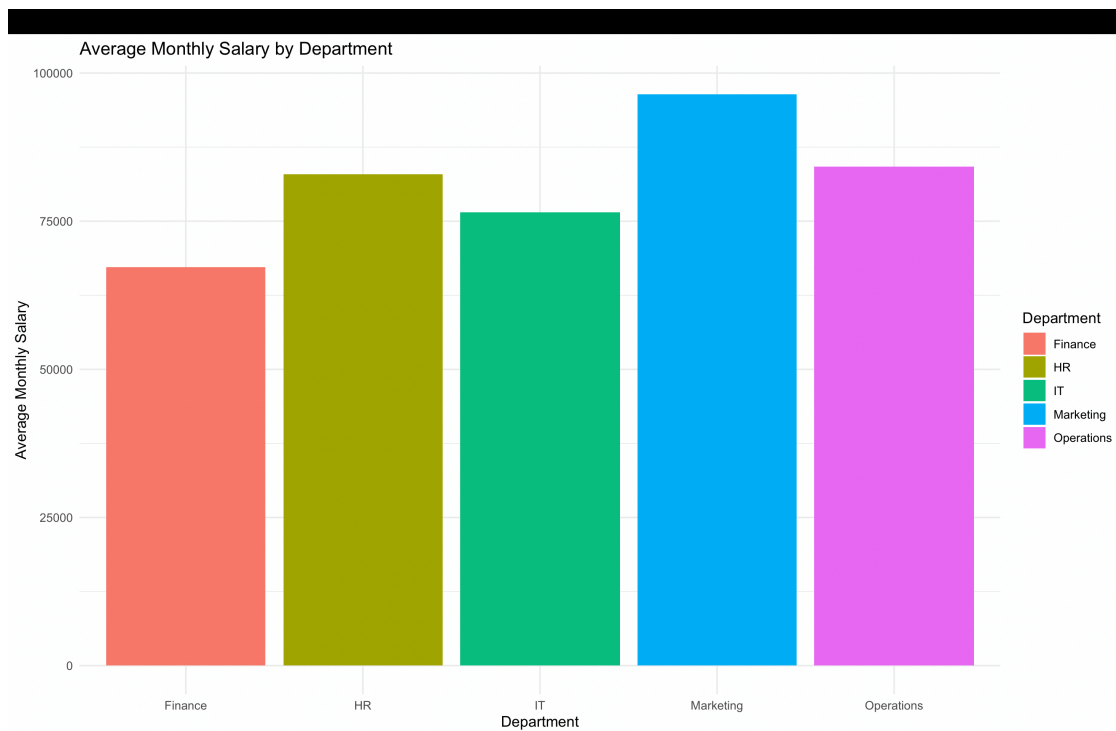
Program:

```
library(ggplot2)  
library(dplyr)
```

```
salary_department <- data %>%  
  group_by(Department) %>%  
  summarise(Average_Salary = mean(Monthly_Salary))
```

```
ggplot(salary_department,  
  aes(x = Department,  
      y = Average_Salary,  
      fill = Department)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Average Monthly Salary by Department",  
    x = "Department",  
    y = "Average Monthly Salary"  
  ) +  
  theme_minimal()
```

Output:



Visualization 2: Grouped Bar Chart

Title: Gender-Based Salary Comparison Across Departments

Program:

```
library(dplyr)

salary_gender <- data %>%
  group_by(Department, Gender) %>%
  summarise(
    Average_Salary = mean(Monthly_Salary),
    .groups = "drop"
  )

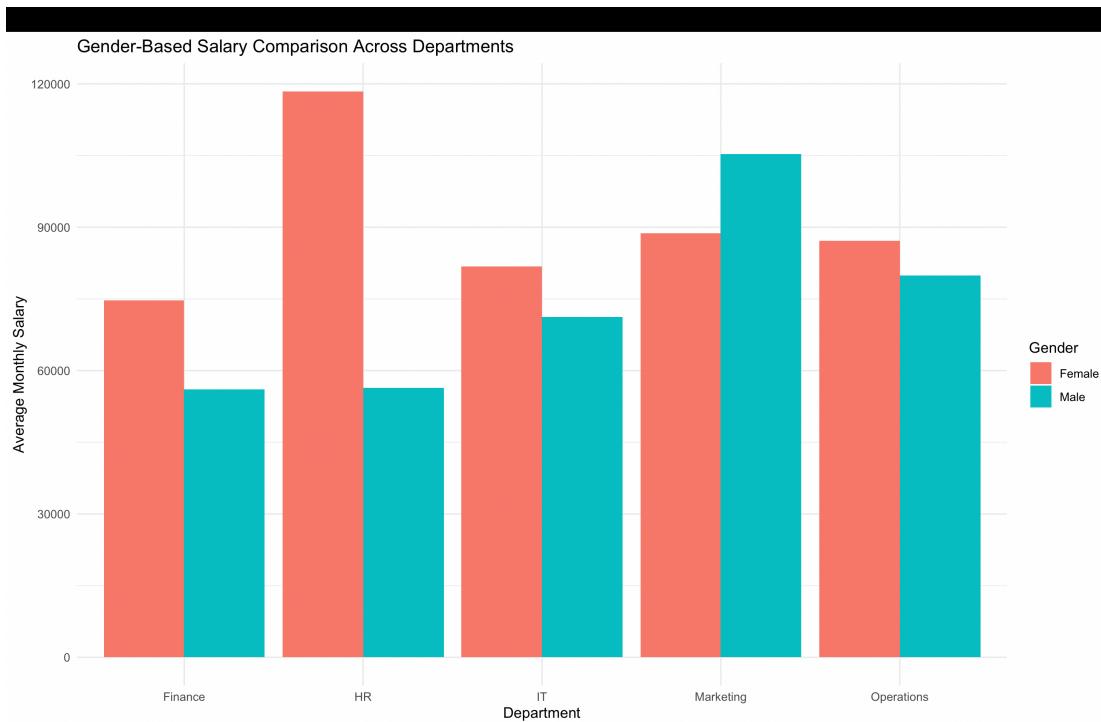
salary_gender

library(ggplot2)

ggplot(salary_gender,
  aes(x = Department,
    y = Average_Salary,
    fill = Gender)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(
    title = "Gender-Based Salary Comparison Across Departments",
    x = "Department",
    y = "Average Monthly Salary",
```

```
fill = "Gender"
) +
theme_minimal()
```

Output:



Visualization 3: Stacked Bar Chart

Topic: Departmental Salary Composition by Gender

Program:

```
salary_gender_total <- data %>%
  group_by(Department, Gender) %>%
  summarise(
    Total_Salary = sum(Monthly_Salary),
    .groups = "drop"
  )

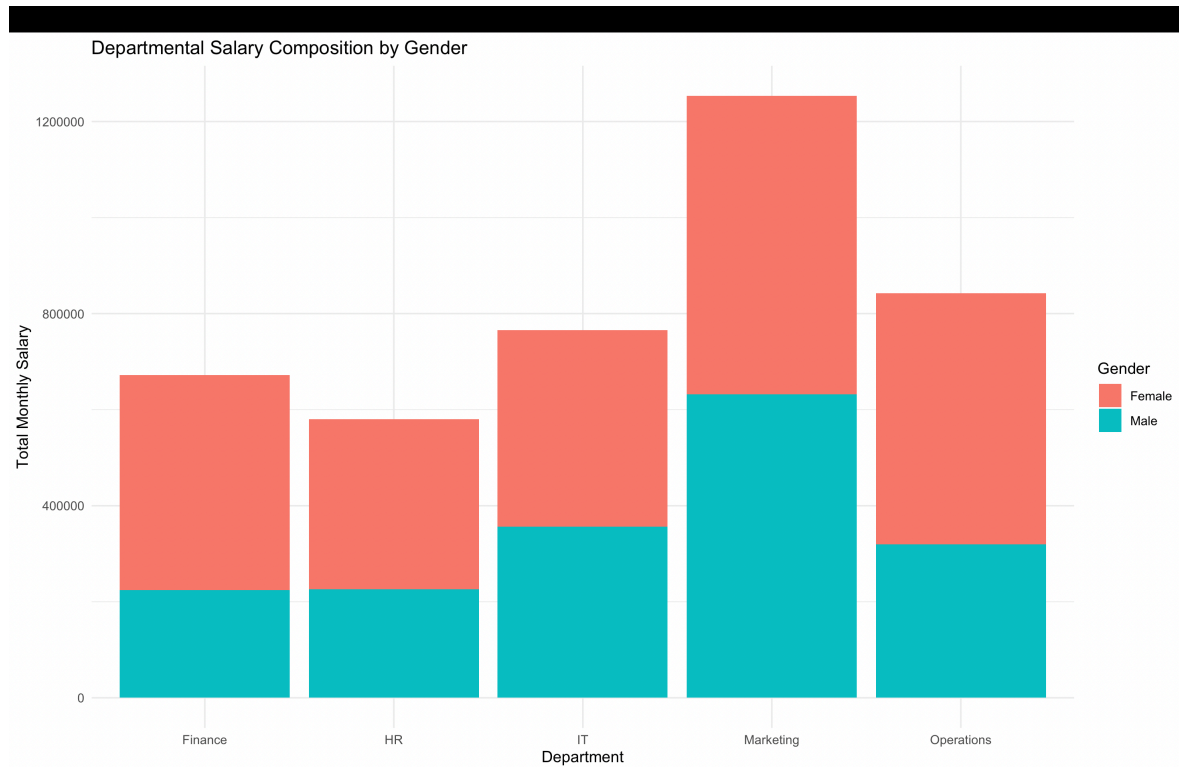
ggplot(salary_gender_total,
  aes(x = Department,
    y = Total_Salary,
    fill = Gender)) +
  geom_bar(
    stat = "identity"
  ) +
  labs(
    title = "Departmental Salary Composition by Gender",
    x = "Department",
```

```

y = "Total Monthly Salary",
fill = "Gender"
) +
theme_minimal()

```

Output:



Visualization 4: Dot Plot

Topic: Individual Salary Amounts Across Departments

Program:

```

ggplot(data,
  aes(x = Monthly_Salary,
    y = Department,
    color = Department)) +
  geom_jitter(
    height = 0.15,
    width = 0,
    size = 3
  ) +
  labs(
    title = "Individual Employee Salary Values Across Departments",
    x = "Monthly Salary",
    y = "Department",
    color = "Department"
  ) +
  theme_minimal()

```



Output:

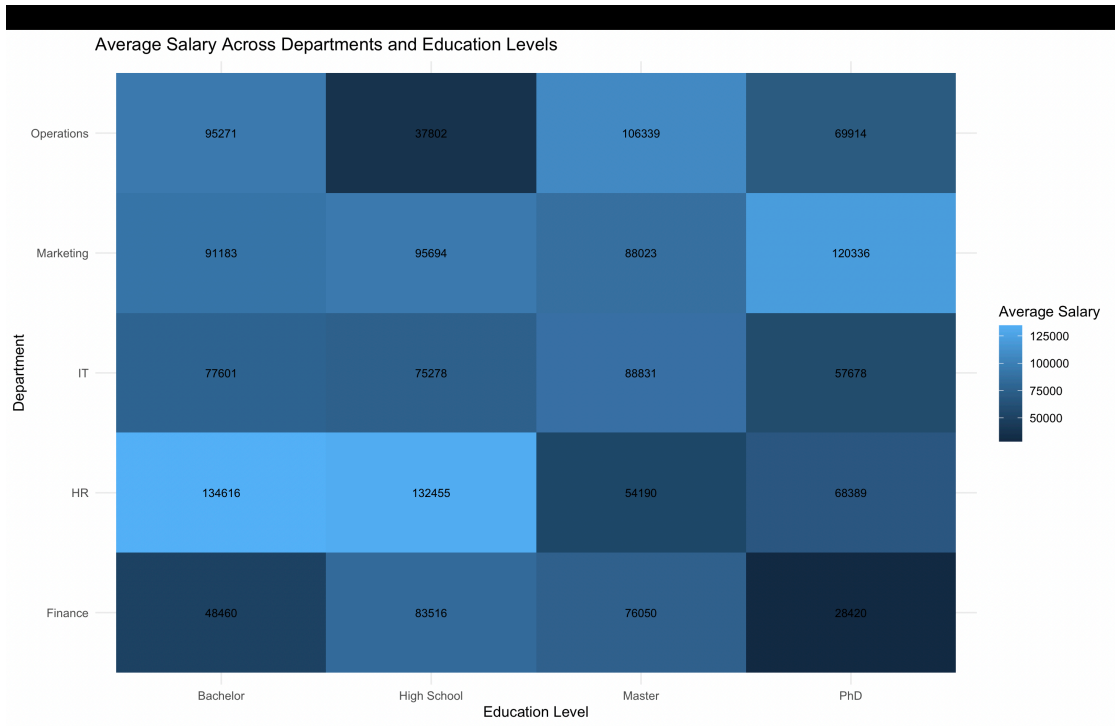
Visualization 5: Heat Map

Topic: Average Salary by Department and Education Level

Program:

```
ggplot(salary_heatmap,
  aes(x = Education_Level,
    y = Department,
    fill = Average_Salary)) +
  geom_tile() +
  geom_text(
    aes(label = round(Average_Salary, 0)),
    color = "black",
    size = 3
  ) +
  labs(
    title = "Average Salary Across Departments and Education Levels",
    x = "Education Level",
    y = "Department",
    fill = "Average Salary"
  ) +
```

theme_minimal()
Output:



3. PROPORTIONS

Visualization 1: Pie Chart
Topic: Employee Distribution by Education Level

Program:

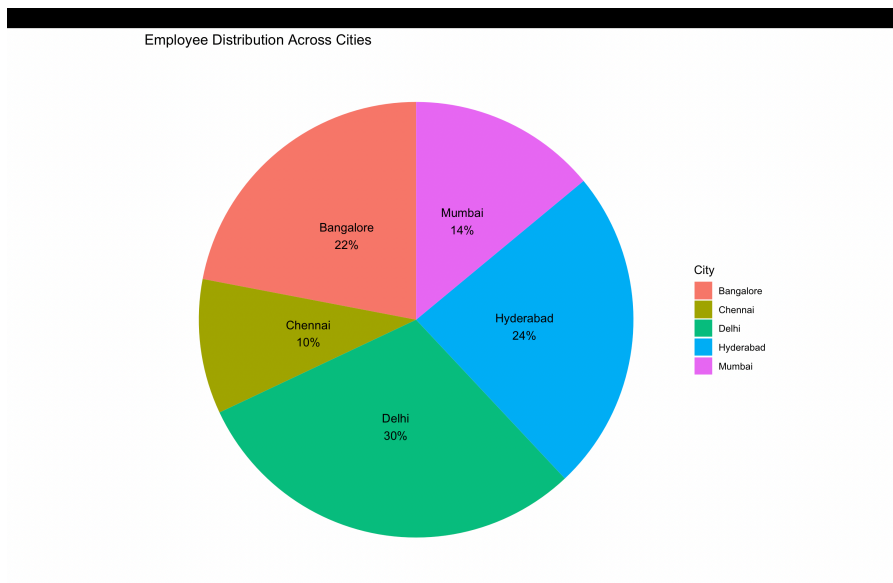
```
city_count <- data %>%  
  count(City) %>%  
  mutate(  
    Percentage = round(n / sum(n) * 100, 1),  
    Label = paste0(City, "\n", Percentage, "%")  
  )  
  
ggplot(city_count,  
  aes(x = "",
```

```

    y = n,
    fill = City)) +
geom_bar(
  stat = "identity",
  width = 1
) +
geom_text(
  aes(label = Label),
  position = position_stack(vjust = 0.5),
  color = "black"
) +
coord_polar("y") +
labs(
  title = "Employee Distribution Across Cities",
  fill = "City"
) +
theme_void()

```

Output:



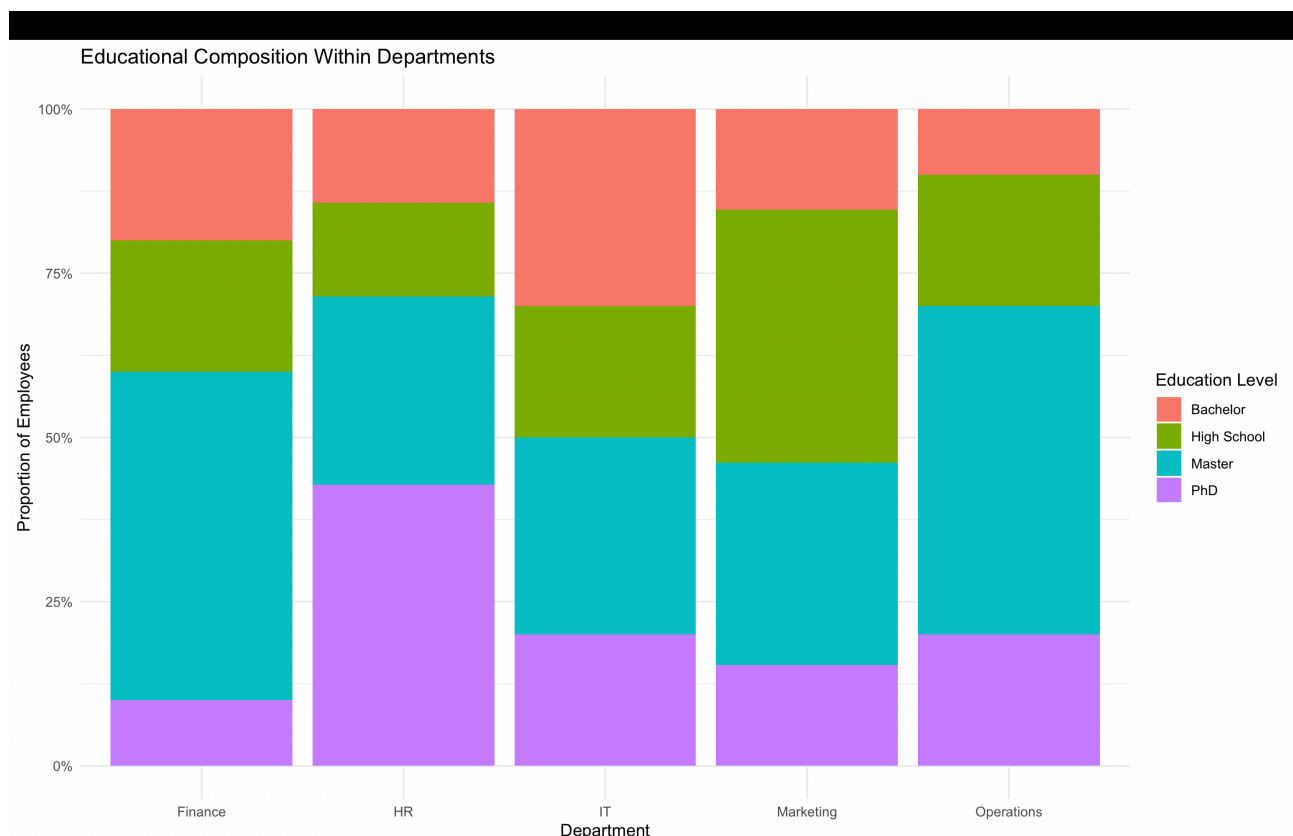
Visualization 2: 100% Stacked Bar Chart

Topic: Educational Composition Within Departments

Program:

```
education_department <- data %>%  
  count(Department, Education_Level)  
  
ggplot(education_department,  
  aes(x = Department,  
    y = n,  
    fill = Education_Level)) +  
  geom_bar(  
    stat = "identity",  
    position = "fill"  
  ) +  
  scale_y_continuous(  
    labels = scales::percent  
  ) +  
  labs(  
    title = "Educational Composition Within Departments",  
    x = "Department",  
    y = "Proportion of Employees",  
    fill = "Education Level"  
  ) +  
  theme_minimal()
```

Output:



Visualization 3: Treemap

Topic: Employee Distribution Across Departments and Cities

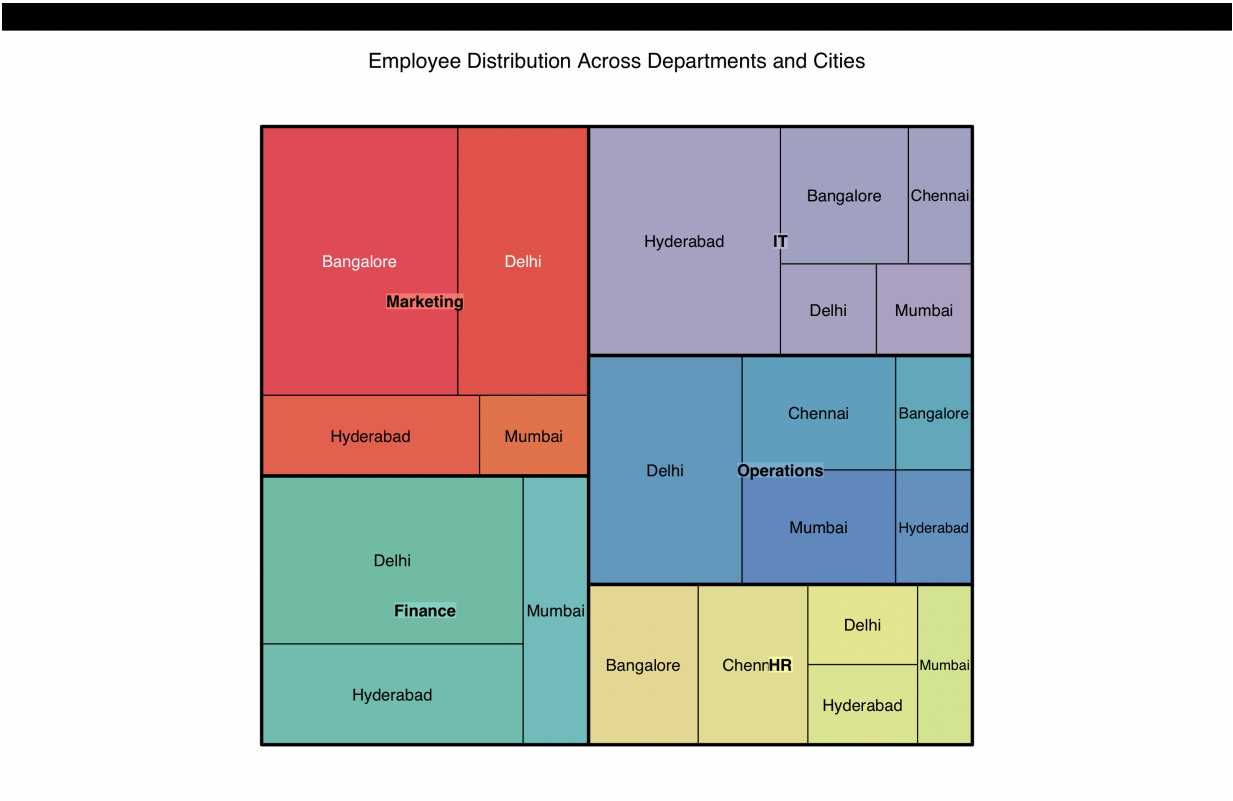
Program:

```
library(treemap)

employee_treemap <- data %>%
  count(Department, City)

treemap(
  employee_treemap,
  index = c("Department", "City"),
  vSize = "n",
  title = "Employee Distribution Across Departments and Cities",
  palette = "Set3"
```

Output:



Visualization 4: Sunburst Chart

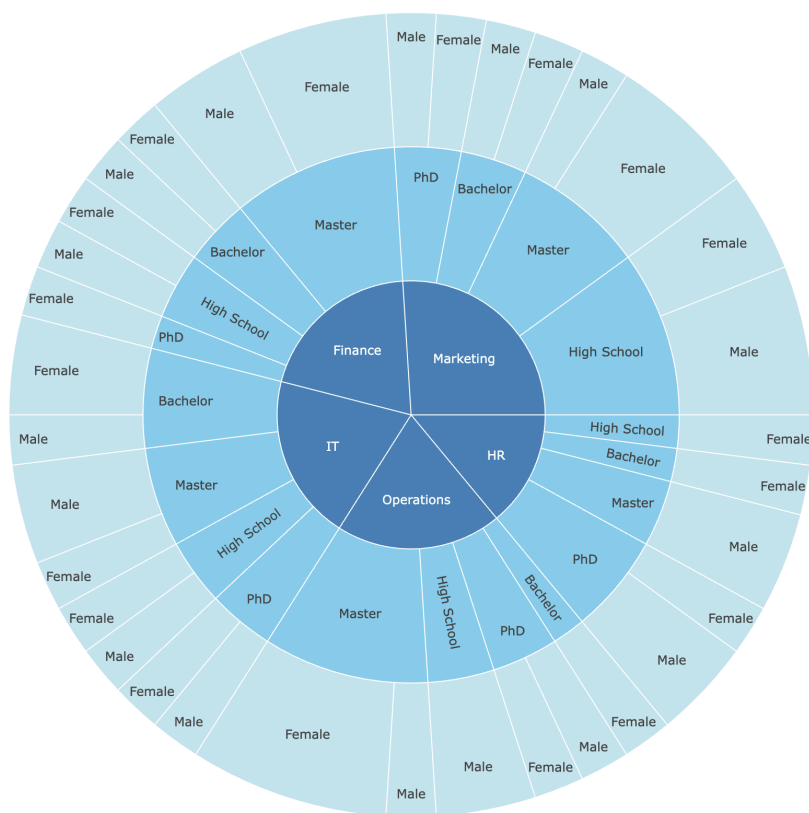
Topic: Hierarchical Employee Composition by Department, Education Level, and Gender

Program:

```
install.packages("plotly")
library(plotly)
sunburst_data <- data %>%
  count(Department, Education_Level, Gender)

plot_ly(
  sunburst_data,
  ids = ~paste(Department, Education_Level, Gender, sep = "-"),
  labels = ~Gender,
  parents = ~paste(Department, Education_Level, sep = "-"),
  values = ~n,
  type = "sunburst"
)
```

Output:



4. DISTRIBUTIONS

Visualization 1: Histogram

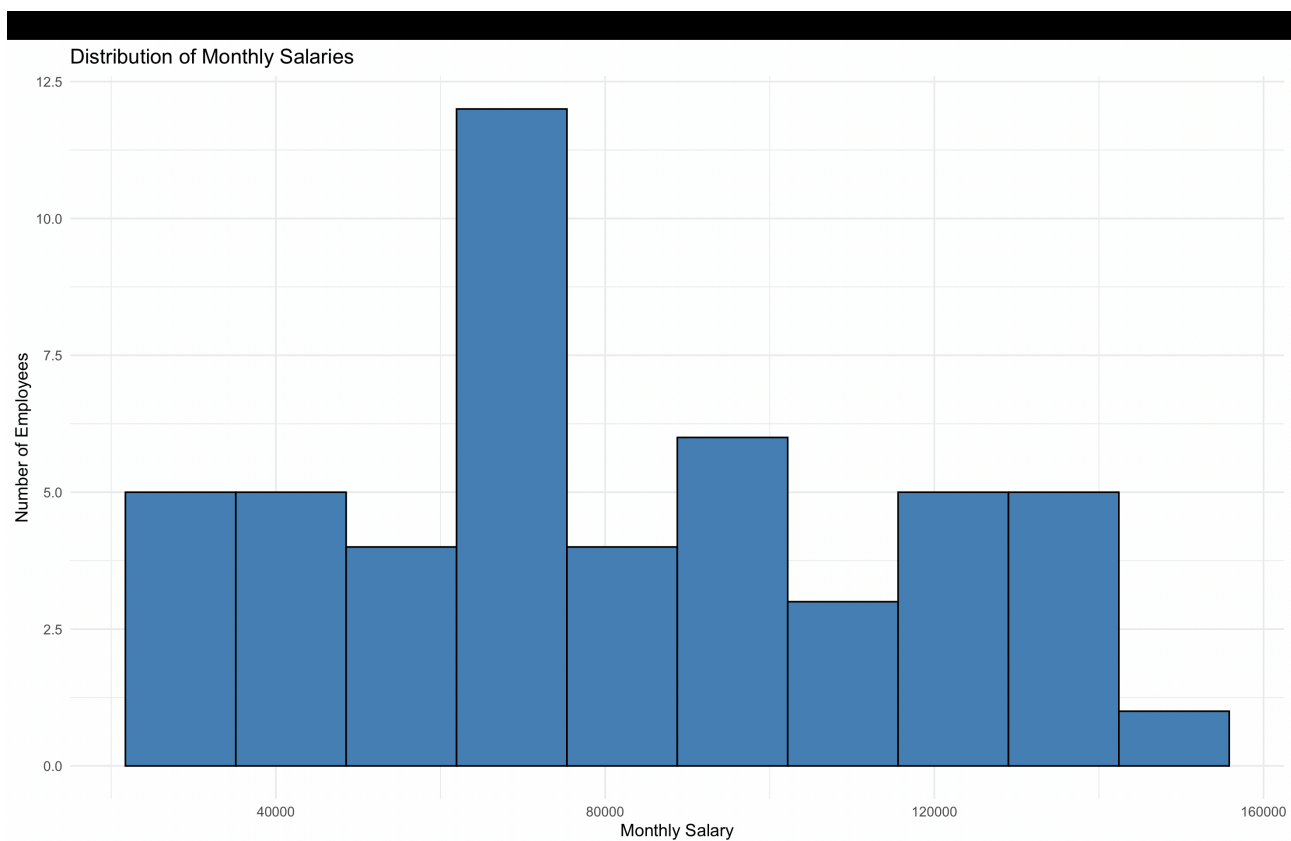
Topic: Distribution of Monthly Salaries

Program:

```
library(ggplot2)

ggplot(data,
       aes(x = Monthly_Salary)) +
  geom_histogram(
    bins = 10,
    fill = "steelblue",
    color = "black"
  ) +
  labs(
    title = "Distribution of Monthly Salaries",
    x = "Monthly Salary",
    y = "Number of Employees"
  ) +
  theme_minimal()
```

Output:



Visualization 2: Density Plot

Topic: Salary Distribution Across Departments

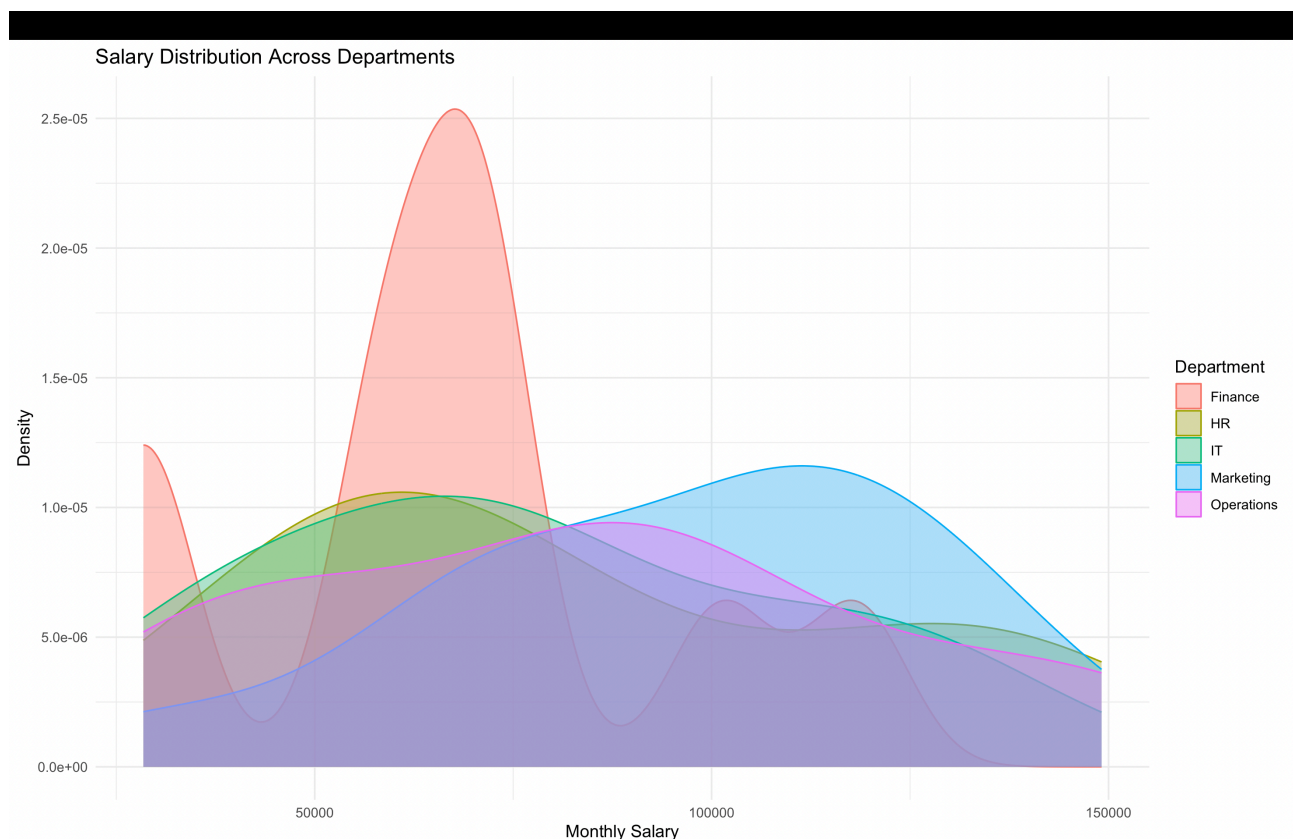
Program:

```
library(ggplot2)

library(ggplot2)

ggplot(data,
       aes(x = Monthly_Salary,
           fill = Department,
           color = Department)) +
  geom_density(
    alpha = 0.4
  ) +
  labs(
    title = "Salary Distribution Across Departments",
    x = "Monthly Salary",
    y = "Density",
    fill = "Department",
    color = "Department"
  ) +
  theme_minimal()
```

Output:



Visualization 3: Box Plot

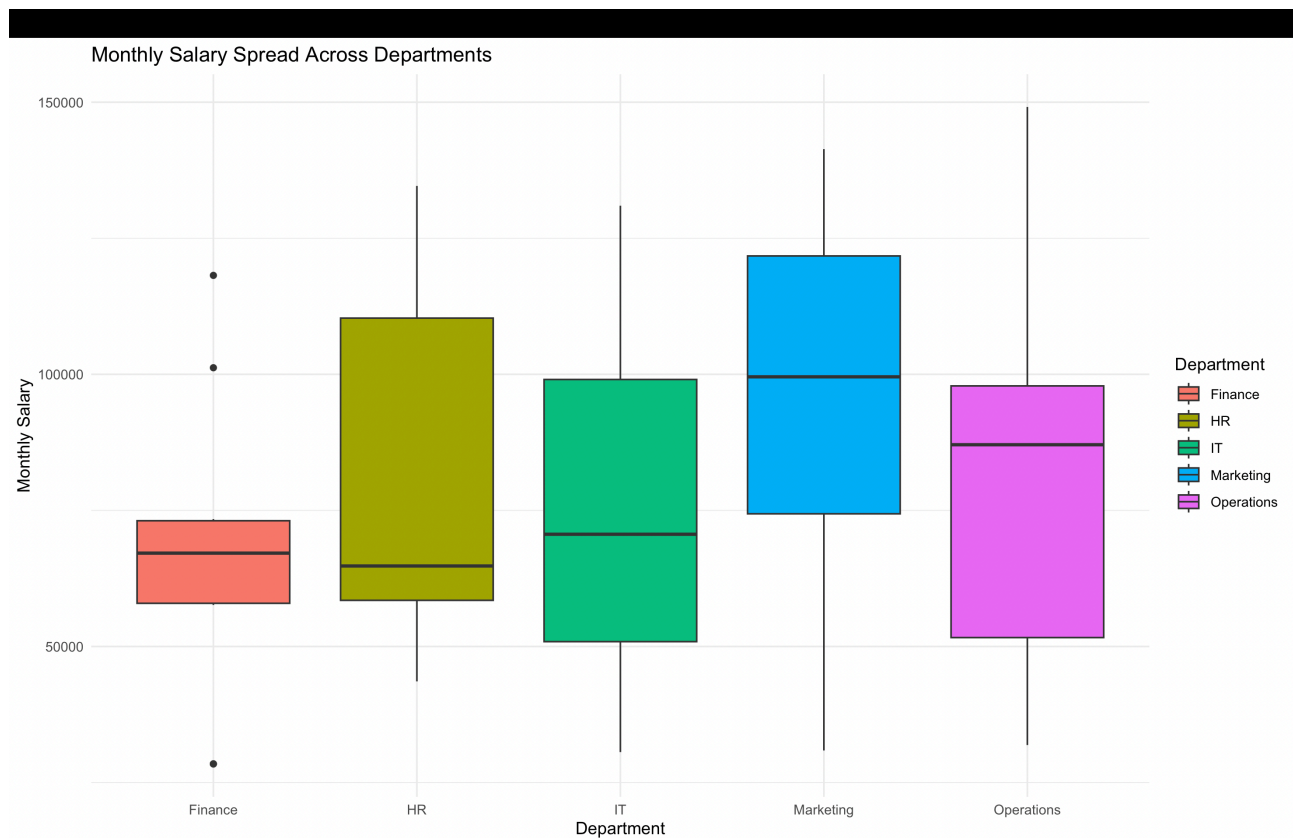
Topic: Monthly Salary Spread Across Departments

Program:

```
library(ggplot2)

ggplot(data,
  aes(x = Department,
      y = Monthly_Salary,
      fill = Department)) +
  geom_boxplot() +
  labs(
    title = "Monthly Salary Spread Across Departments",
    x = "Department",
    y = "Monthly Salary"
  ) +
  theme_minimal()
```

Output:



Visualization 4: Violin Plot

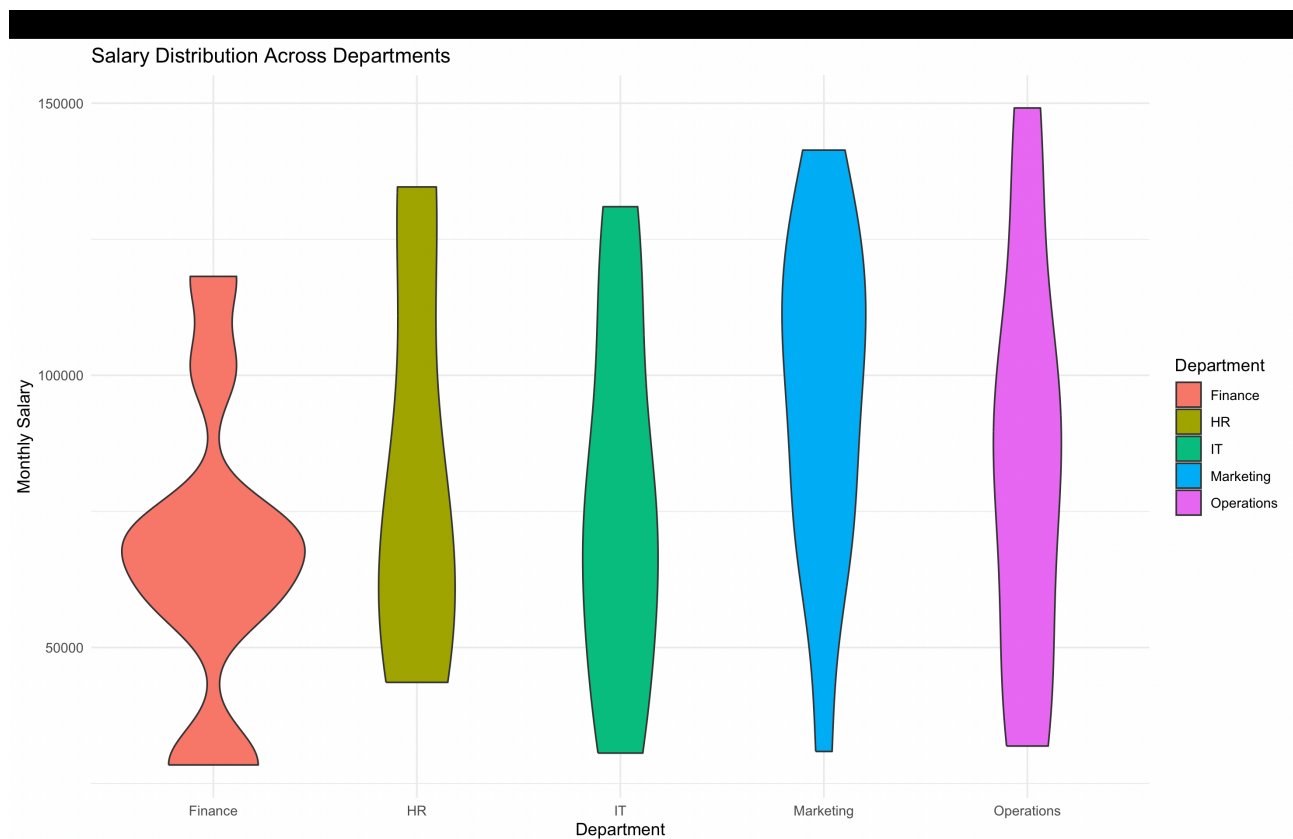
Topic: Salary Distribution Across Departments

Program:

```
library(ggplot2)

ggplot(data,
       aes(x = Department,
           y = Monthly_Salary,
           fill = Department)) +
  geom_violin() +
  labs(
    title = "Salary Distribution Across Departments",
    x = "Department",
    y = "Monthly Salary"
  ) +
  theme_minimal()
```

Output:



Visualization 5: Ridgeline Plot

Topic: Salary Distribution Across Departments

Program:

```
install.packages("ggridges")
library(ggplot2)
library(ggridges)

ggplot(data,
       aes(x = Monthly_Salary,
           y = Department,
           fill = Department)) +
  geom_density_ridges(alpha = 0.7) +
  labs(
    title = "Salary Distribution Across Departments",
    x = "Monthly Salary",
    y = "Department"
  ) +
  theme_minimal()
```

Output:

